

Protein Synthesis

Protein synthesis has two stages. Transcription occurs in the nucleus, where the DNA sequence of a gene is copied into messenger RNA (mRNA). The mRNA then leaves the nucleus through a nuclear pore.

Translation takes place at the ribosome. Transfer RNA (tRNA) molecules deliver amino acids to the ribosome, where they are joined in the order specified by the mRNA codons to build a protein chain.

Cell Organelles

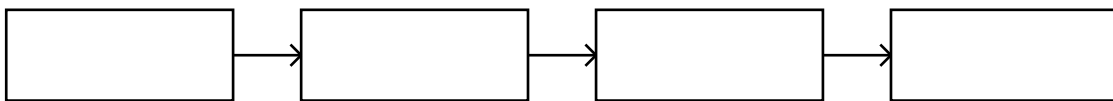
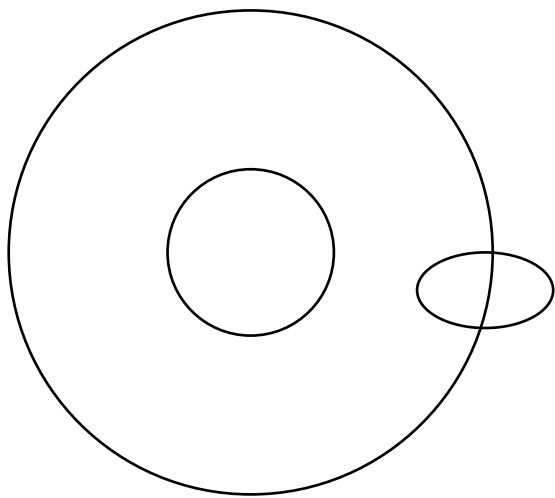
Table 1 lists the main organelles and their functions.

Organelle	Function
Nucleus	Stores DNA and controls the cell
Mitochondrion	Produces ATP through cellular respiration
Ribosome	Builds proteins from amino acids
Chloroplast	Captures light for photosynthesis
Cell membrane	Controls what enters and leaves the cell

Cell Diagrams

Figure 1: Animal cell diagram. Labeled parts: nucleus, mitochondria, ribosome, and cell membrane.

Figure 2: Stages of mitosis in order. The flowchart reads: Prophase, then Metaphase, then Anaphase, then Telophase.



Mitochondrion Figure

Figure 3: Mitochondrion structure (see diagram).

Mitochondrion

